

Justification of the Symbolic Recurrence Method as a Replacement for Lagrange Interpolation

Setup

Fix rational numbers $j_1, j_2, j_3, \dots$ (in the application, these are derived from the coefficients $\tau(p)$ of the discriminant cusp form, but nothing below depends on their origin). Define a sequence $(h_n)_{n \geq 0}$ by the recurrence

$$h_0 = 1, \quad h_n = \frac{1}{n} \left(j_n + \sum_{r=1}^{n-1} j_r h_{n-r} \right), \quad n \geq 1. \quad (1)$$

The *deformed* computation prepends a parameter to the j -sequence: for $c_0 \in \mathbb{Q}$, apply (1) to the input sequence $(c_0, j_1, j_2, \dots)$, i.e. with j_1 replaced by c_0 and the original j_r shifted to position $r + 1$. Write $h_n(c_0)$ for the resulting value, and define

$$P_n(c_0) = n! h_n(c_0).$$

The original numerical method recovers P_n by evaluating $h_n(c_0)$ at the integer nodes $c_0 = 1, 2, \dots, 2n + 4$ and applying Lagrange interpolation. The symbolic method instead runs (1) once over the polynomial ring $\mathbb{Q}[c]$ with input sequence $(c, j_1, j_2, \dots)$.

The methods compute the same polynomial

Lemma 1 (Evaluation commutes with the recurrence). *Let $H_n \in \mathbb{Q}[c]$ denote the output of recurrence (1) applied over $\mathbb{Q}[c]$ to the input $(c, j_1, j_2, \dots)$. Then for every $c_0 \in \mathbb{Q}$,*

$$H_n(c_0) = h_n(c_0).$$

Proof. For fixed $c_0 \in \mathbb{Q}$, the evaluation map $\text{ev}_{c_0}: \mathbb{Q}[c] \rightarrow \mathbb{Q}$, $f \mapsto f(c_0)$, is a ring homomorphism. The recurrence (1) is built exclusively from ring

operations: addition, multiplication, and multiplication by the fixed scalars $1/n \in \mathbb{Q}$. A ring homomorphism (which here is moreover $\mathbb{Q}$ -linear) commutes with every expression formed from such operations. Since ev_{c_0} sends the symbolic input sequence $(c, j_1, j_2, \dots)$ to the numerical input sequence $(c_0, j_1, j_2, \dots)$, induction on n gives $\text{ev}_{c_0}(H_n) = h_n(c_0)$ for all n . $\square$

Note that the argument uses no property of the data $j_1, j_2, \dots$ whatsoever; it is an identity in the inputs, valid for arbitrary rational (indeed, arbitrary commutative-ring) values.

Lemma 2 (Degree bound). *$\deg_c H_n = n$, with leading term $c^n/n!$.*

Proof. By induction. $H_0 = 1$ has degree 0. In the deformed input, the parameter c occupies position 1; every other input is a constant. In (1), the term $j_r h_{n-r}$ contributes c -degree at most $1 + \deg H_{n-1}$ (when $r = 1$, since the deformed $j_1 = c$) and at most $\deg H_{n-2}$ otherwise, while the lone term j_n is constant in c for $n \geq 2$. Hence $\deg H_n \leq 1 + \deg H_{n-1}$, and the unique contribution attaining this bound is $\frac{1}{n} c H_{n-1}$. Its leading coefficient is $\frac{1}{n} \cdot \frac{1}{(n-1)!} = \frac{1}{n!}$, which is nonzero, so the degree is exactly n . $\square$

Proposition 1 (Equivalence). *For each $n \geq 2$, the polynomial $n! H_n \in \mathbb{Q}[c]$ produced by the symbolic method equals the polynomial produced by Lagrange interpolation of the values $P_n(c_0) = n! h_n(c_0)$ at the nodes $c_0 = 1, 2, \dots, 2n+4$.*

Proof. Let L_n be the Lagrange interpolant: the unique polynomial of degree $\leq 2n+3$ satisfying $L_n(c_0) = P_n(c_0)$ at all $2n+4$ nodes. By Lemma 1, the polynomial $n! H_n$ also satisfies $n! H_n(c_0) = n! h_n(c_0) = P_n(c_0)$ at every node, and by Lemma 2 its degree is $n \leq 2n+3$. Uniqueness of the interpolant of degree $\leq 2n+3$ through $2n+4$ points forces $L_n = n! H_n$. $\square$

Corollary 1. *The two methods agree not only at the interpolation nodes but at every $c_0 \in \mathbb{Q}$ (indeed, as elements of $\mathbb{Q}[c]$, coefficient by coefficient). In particular, the symbolic method introduces no modeling assumption about the data: it defers the evaluation step of the original computation rather than altering it.*

Remark 1. Proposition 1 also explains an empirical observation of the numerical method: the interpolants were always found to have degree exactly n . By Lemma 2 this is a structural property of the recurrence — a consequence of the parameter entering linearly in position 1 — and holds for arbitrary input data. The data determine the lower-order coefficients of H_n , not its degree.

Remark 2 (Falsifiability). The equivalence can be tested independently of the proof: evaluate $n! H_n$ at points that were never interpolation nodes (e.g. negative integers or non-integers such as $c_0 = -7$ or $3/2$) and compare with the direct deformed computation $P_n(c_0)$; or compare the two polynomials coefficient by coefficient with exact arithmetic. Agreement at non-node points cannot be an artifact of fitting to the nodes.